

v3 was tuned on the sealed set; re-deriving its thresholds honestly costs four gates

Results

Three arms, one scorer, the eight sealed data/2-eval corpora, the two policies this project declared before any of them existed. v3 was scored **through its own packaged runtime**; v2 likewise; v4 was built during this run.

metric	v2	v3	v4
macro F2 (<i>headline ranker</i>)	0.6614	0.6657	0.6641
micro F1	0.7255	0.7533	0.7862
priority macro F0.5 (<i>contra-view ranker</i>)	0.7434	0.7587	0.8031
macro F0.5	0.6183	0.6253	0.6403
recall, macro over catalogue	0.6938	0.6887	0.6655
precision, macro over catalogue	0.6182	0.6258	0.6461
worst measurable priority recall	0.7221	0.7221	0.6410
prediction rate	0.8192	0.8181	0.8168
document recall / precision / specificity	0.7975 / 0.8956 / 0.8792	<i>identical</i>	<i>identical</i>
one-core p95	3.9164 ms	3.9164 ms	3.9696 ms

Document-level metrics are identical across all three to four decimal places, and this was predicted before measuring rather than discovered: `quiet_runtime.predict` consults the gate **before** scoring any head, and all three arms carry byte-identical `gate_weights` and `gate threshold`. Only tag emission can move.

Both declared policies, unchanged

policy	v2	v3	v4
headline — per-tag recall ≥ 0.90 on <code>ci_lower</code>	blocked, 29/55	blocked, 29/55	blocked, 25/55
precision view — doc precision/specificity/recall + tag recall ≥ 0.75	blocked, 45/55	blocked, 45/55	blocked, 44/55
one-core p95 ≤ 5 ms	PASS	PASS	PASS

NO FEASIBLE ARM under both, for all three. Nothing was promoted and nothing was packaged.

What v4 changed, and what it cost

The corrected cap moved **five** of 58 thresholds. On the calibration curve:

tag	v2 threshold → v4	v2 P / R / F0.5	v4 P / R / F0.5	v2 → v4 firings
27 date of death	−4.13 → 9.56	0.4145 / 0.9994 / 0.4694	0.9862 / 0.9703 / 0.9830	48,834 → 18,895
44 military ID	−3.40 → 10.03	0.2150 / 0.9992 / 0.2550	0.9364 / 0.8262 / 0.9121	51,997 → 9,321
47 password	−1.40 → 3.11	0.1094 / 0.9732 / 0.1330	0.8380 / 0.7772 / 0.8251	11,270 → 1,225
49 phone number	13.13 → 13.18	0.9589 / 0.9228 / 0.9514	0.9602 / 0.9215 / 0.9522	20,266 → 20,204
57 zip code	9.87 → 12.33	0.8547 / 0.9548 / 0.8730	0.9518 / 0.8722 / 0.9347	20,066 → 16,372

On the sealed corpora that improvement is paid for in recall, and four priority gates flip:

tag @ corpus	support	v2 recall (ci_low)	v4 recall (ci_low)
military ID @ 20000_pii_holdout	4,327	1.0000 (1.0000)	0.8343 (0.8230)
military ID @ 38937_openpii_eval	13,763	0.9995 (0.9991)	0.7701 (0.7628)
password @ 10626_ai4privacy	1,245	0.9855 (0.9783)	0.7173 (0.6916)
password @ 20000_pii_holdout	232	0.9741 (0.9526)	0.8793 (0.8362)

All four are PASS→FAIL. No pair moved the other way.

The finding that matters most

v3 set tag 44's threshold to **7.4621** by sweeping the sealed evaluation corpora. The training-only procedure, run blind, chose **10.0263** for the same tag. Both raise it from −3.40; the difference is where they stop:

	tag 44 recall @ 20000_pii_holdout	@ 38937_openpii	gates held
v2 (−3.40)	1.0000 (ci_low 1.0000)	0.9995 (0.9991)	both
v3 (7.4621, swept on eval)	0.9609 (0.955)	0.9372 (0.933)	both
v4 (10.0263, training only)	0.8343 (0.823)	0.7701 (0.763)	neither

v3 landed on the one part of the range that captures most of the precision gain *and* stays above the sealed 0.90 gate. That is not luck and it is not skill — it is what selecting against the sealed set buys, and it is exactly the advantage the sealed set exists to deny. The training carve cannot see where the gate sits on 20000_pii_holdout, so it overshoots.

TL;DR

- **v3's two thresholds were selected on the sealed evaluation corpora.** Its own config.json says so: a per-tag F2 sweep on "a held-in half of the eight data/2-eval corpora". The script that did it was never checked in, so the split cannot be reproduced.

- **v3's artifact claims check out otherwise.** `gate_weights` and `tag_weights` are byte-identical to v2; exactly 2 of 58 `tag_thresholds` differ, at indices 44 and 51, as its card says. The `weights.npz` size change is a re-save, not a content change.
- **Re-scored sealed through its own runtime, v3 is real but small:** macro F2 0.6614 → 0.6657, micro F1 0.7255 → 0.7533, and it loses no gates. 96 sampled documents re-scored through the bundle's own `predict()` matched the cached path exactly.
- **The root cause was never the two tags.** `group_recall_cap` lets any source group with ≥ 20 positives set the cap, and at a 0.7556 floor that is a 5th order statistic. One 31-positive slice of `148775_pii2_train` (AUC 0.684) dictated tag 44's threshold for 10,556 positives, against sibling caps of +10.38 and +11.57.
- **Fixing that properly moves five thresholds, not two** — including date of death, password and zip code, which v3 never touched, and which were firing 48,834, 11,270 and 20,066 times on the calibration curve at precision 0.41, 0.11 and 0.85.
- **v4 wins every precision-priced aggregate and loses the headline anyway:** priority macro F0.5 0.7434 → 0.8031, micro F1 → 0.7862, but macro F2 stays flat (0.6641) because F2 weights recall four times, and four priority gates flip PASS→FAIL. **29/55 → 25/55.**
- **Tag 51 was not fixed**, and neither was it fixable this way: its cap-setting group clears the ten-tail-events bar (43 positives → 10.5 in the tail). It is also **not measurable on any sealed corpus**, so v3's headline claim for it (F2 0.026 → 0.444) cannot be confirmed or refuted here at all.
- **Nothing promoted, nothing packaged.** Under the declared headline policy v4 is worse than the artifact it corrects. Packaging it would ship a bundle whose card had to explain why the honest model scores lower.

Date	2026-08-26
Project	projects/pii-head-to-head-v1
Run	26-08-26-4
Request	"Re-measure & report v3", then "also re-derive thresholds reproducibly"
Scope	3 arms × 8 sealed corpora; 5 of 58 thresholds re-derived on the training curve
CPU budget	1 core (the default) for every latency; training curve selection unbudgeted
Outcome	v3 audited and confirmed contaminated; v4 built, measured, and not shipped
Commands	26-08-26-4_v3-audit-and-v4-thresholds.commands.txt (29, failures included)
Approval	approvals/threshold-rederivation-v4.json
Tracking	sqlite:///projects/pii-head-to-head-v1/mlflow.db, experiment pii-head-to-head-v1, parent run 7cc1e84f — 1 parent + 3 model runs + 24 model×corpus runs. No model registered and no alias moved: nothing here is promotable
Workbook	26-08-26_Experiment-Log.xlsx — 153 log rows (129 carried forward + 24), 1,135 per-tag rows

Model scoreboard

Model	Dataset	F1 @5	F1 @3	F1 @1	Macro F1	Docs/ sec	ms/ doc
pii-cascade-balanced-v2	10360_betterdataai_ner_silver_eval_10.36k				0.289	255.3	3.9
pii-cascade-balanced-v2	10626_ai4privacy_pii_masking_eval_10.63k				0.742	255.3	3.9
pii-cascade-balanced-v2	20000_pii_holdout_20.00k				0.704	255.3	3.9
pii-cascade-balanced-v2	30000_pii2_eval_25.15k				0.757	255.3	3.9
pii-cascade-balanced-v2	38937_openpii_pii_eval_38.94k				0.658	255.3	3.9
pii-cascade-balanced-v2	4000_datax-dualjudge-evalset-1.32k					255.3	3.9
pii-cascade-balanced-v2	5617_nemotron_eval_5.36k					255.3	3.9
pii-cascade-balanced-v2	6589_govdocs2-dualjudge-eval20-3.53k					255.3	3.9
pii-cascade-balanced-v3	10360_betterdataai_ner_silver_eval_10.36k				0.289	255.3	3.9
pii-cascade-balanced-v3	10626_ai4privacy_pii_masking_eval_10.63k				0.742	255.3	3.9
pii-cascade-balanced-v3	20000_pii_holdout_20.00k				0.720	255.3	3.9
pii-cascade-balanced-v3	30000_pii2_eval_25.15k				0.760	255.3	3.9
pii-cascade-balanced-v3	38937_openpii_pii_eval_38.94k				0.669	255.3	3.9
pii-cascade-balanced-v3	4000_datax-dualjudge-evalset-1.32k					255.3	3.9
pii-cascade-balanced-v3	5617_nemotron_eval_5.36k					255.3	3.9
pii-cascade-balanced-v3	6589_govdocs2-dualjudge-eval20-3.53k					255.3	3.9
pii-cascade-v4-corrected-cap	10360_betterdataai_ner_silver_eval_10.36k				0.290	251.9	4.0
pii-cascade-v4-corrected-cap	10626_ai4privacy_pii_masking_eval_10.63k				0.764	251.9	4.0
pii-cascade-v4-corrected-cap	20000_pii_holdout_20.00k				0.727	251.9	4.0
pii-cascade-v4-corrected-cap	30000_pii2_eval_25.15k				0.764	251.9	4.0
pii-cascade-v4-corrected-cap	38937_openpii_pii_eval_38.94k				0.674	251.9	4.0

Model	Dataset	F1 @5	F1 @3	F1 @1	Macro F1	Docs/ sec	ms/ doc
pii-cascade-v4-corrected-cap	4000_datax-dualjudge-evalset-1.32k					251.9	4.0
pii-cascade-v4-corrected-cap	5617_nemotron_eval_5.36k					251.9	4.0
pii-cascade-v4-corrected-cap	6589_govdocs2-dualjudge-eval20-3.53k					251.9	4.0

Measured 2026-08-26 · n=10,360, n=10,626, n=20,000, n=30,000, n=38,937, n=4,000, n=5,617, n=6,589.